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TANDEM REPORTS TXP BENCHMARKS

Summary: New ET1 benchmark data from Tandem show the TXP ranging over .2-9.6 tps/CPU, and, according to Tandem, show a significant edge over IBM's PF2 in cost per tps.

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Tandem recently shared with FTSN ome of the results of a series of ET1 enchmarks run on a 4-processor TXP ystem. ET1 is the "debit/credit" ransaction, as logically defined in the rticle by "anon et al" in the 1 April issue of Datamation, but with 1/5 the database size, as follows (each record = 100 bytes):

2 million account records
2000 teller records
200 branch record

The physical configuration consisted of a TXP system with four 4MB CPUs, 6 or 7 disks (two 128MB 4110s, two 300MB 4104s, and either two or three 168MB V8's), and one 6104 communications controller operating under X.25 at an aggregate of 76 Kbaud. Each teller terminal represents a separate virtual circuit. The system under test was driven by a separate, 2-TXP "load generator" system.

Several versions of the benchmark program were tried, including one in which each terminal was served by a separate process, and neither TMF nor Pathway were employed ("Version 2"); another ("Version 5") did utilize the standard Pathway requester/server model, but no TMF; and still another ("Version 6") employed both Pathway and TMF working with the new Guardian release B00 and new DP2 disk process. All non-TMF versions used the seven-disk configuration; with TMF, only six disks could be effectively used.

Results were as follows:

	Version	TPS/ processor	Cost/TPS
NO TMF	V.2	9.56	$33.7K
PATHWAY	V.5	7.60	$39.2K
TMF/PATHWAY/9	V.6 "new"	4.70	$58 K
''	V.6 "old"	3.17	$80.6K

where the "old" V.6 used the Guardian A06 release with DP1. Average response time of under 1 second is maintained in all cases.

Costs are calculated as suggested in the anon et. al. article, by summing the following elements:

Hardware purchase price
5-year maintenance costs
Software initial license fees
Software 5-year maintenance fees

Dennis McEvoy, Tandem's vice president of software, told FTSN that these results proved a number of points. Relative to IBM's TPF2, Tandem believes its V.2 and V.5 cost/TPS are substantially lower, even though both cases are coded in Cobol as opposed to assembly language, mandated by TPF2. V.6, which utilizes TMF and Pathway, is much richer in functionality, and should be compared to an IMS or IMS/Fastpath case, in which case again Tandem has a cost/TPS edge.

(Tandem did not do the benchmarks on TPF2 or IMS, but based on data from sources believed to be expert on these systems, Tandem believes its price-performance claims are justified).

Another point is that, accepting previous results which showed that the TXP runs about 2.5 times faster than the NS-II, McEvoy feels that the new data points to a range of 1.9-3.8 tps/CPU for the NS-II (assuming the new B00/DP2 software); this is much higher than the 1.0-1.1 tps/CPU reported by Stratus (FTSN-32 p.6).

Thirdly, McEvoy thinks these results prove that a 16-CPU TXP system can do 100 tps, and that a 10-system FOX network should therefore be able to do the magic "1000 tps".

The benchmark work was done by a group headed by Harald Sammer in Frankfurt, W. Germany. The group is informally known as the "1000 tps" group; its mission is to find ways in which Tandem standard hardware and software can be modified to reach high transaction throughput.

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